

Pair-Conditioned Verification of Literature-Mined Species Interactions

Anonymous ACL submission

Abstract

Species interaction databases (who eats, infects or pollinates whom) are partly built by mining the literature. Candidates are proposed when a passage names two taxa near an interaction term, and the obvious filter asks whether the passage describes an interaction. A passage naming several taxa may support only some of its pairs, yet this filter gives them all the same answer. In our benchmark, passages naming three or more taxa hold 62% of the interacting pairs. Keeping a 110M-parameter encoder and its LLM-labelled training data fixed, we vary the input: the passage alone, the passage with the candidate pair, or the passage with the pair and its interaction term. On 437 expert-graded candidates, adding the pair raises the area under the precision-recall curve (AUPRC) from 0.867 to 0.942, and the interaction term adds no significant further gain. The gain is largest on passages naming three or more taxa. It replicates on the public BioRED corpus and in zero-shot Qwen models from 1.7B to 122B parameters, whose answers improve with size much more when asked about the pair. Asked only about the passage, the 32B model that labelled our training data ranks candidates below the 110M encoder given the pair.

1 Introduction

Databases such as GloBI (Poelen et al., 2014) record which species eat, infect, parasitise or pollinate which, and like many biomedical databases (Hirschman et al., 2012; Lee et al., 2018) they are populated in part by text mining (Dimitrova et al., 2020). A retrieval system proposes candidate triples (s_1, r, s_2) , two organisms and an interaction term, together with the passage that allegedly supports them, and something must decide which candidates are worth a curator’s attention. A natural choice is a sentence-level classifier that reads the passage and decides whether it describes a biotic interaction.

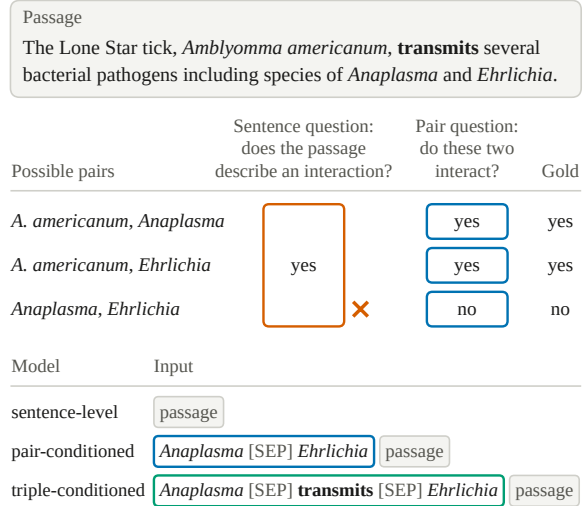


Figure 1: One passage, several candidate pairs. Retrieval proposes a candidate when a passage names two organisms near an interaction word (bold). The sentence question gets one answer for the whole passage, so even a perfect answer accepts the third pair, two bacterial genera merely listed together. The third pair is a real benchmark item (TEST299, graded negative) that the sentence-level model accepts and both conditioned models reject at their block-held-out thresholds; the first two pairs are illustrative. Bottom: model inputs.

This classifier gives one answer per passage, whereas co-occurrence retrieval proposes candidates pair by pair: a passage naming two taxa near an interaction term yields a candidate whether or not it asserts that interaction between those two. Every candidate therefore already contains the wording a sentence-level classifier responds to: in the retrieval pool we study (§3.1), all 175,588 candidates carry an interaction term, and a lexical scan for GloBI interaction vocabulary fires on 99.83% of their passages. The decision that matters is narrower: does the passage assert an interaction between these two taxa? A passage that names several organisms can support one pair and not another, and a classifier that sees only the passage gives the

same answer whichever pair the candidate names; even a perfect answer to its question accepts the unsupported pair (Figure 1).

Ruch et al. (2024), who built the retrieval system, graded 100 random BiotXplorer triples, found that the system identified the interacting species with a precision of 31%, and named the main cause of error: “instances where passages listed multiple species, which can be automatically filtered out.” On our benchmark, passages naming three or more taxa hold 156 of the 251 interacting pairs (62%), so filtering them out would discard most of what the pipeline exists to find. We keep those passages and judge each candidate by its pair.

To test this, we train the same 110M-parameter encoder three times on the same corpus, labelled by a 32B language model, reading the passage alone (*sentence-level*), the passage with the candidate pair (*pair-conditioned*), or the passage with the whole candidate triple (*triple-conditioned*), and compare them on 437 expert-graded candidates. Everything except the input is held fixed. Telling the model which pair it is judging raises the area under the precision-recall curve (AUPRC, which summarises precision over all recall levels; a random ranking scores the share of positives) from 0.867 to 0.942 and F1 from 0.790 to 0.882 (McNemar $p = 3.8 \times 10^{-7}$), and at equal review workload keeps F1 ahead or level at every share of candidates accepted except 93% to 96%, where it trails by at most 0.003; adding the relation term helps only at strict operating points. On BioRED (Luo et al., 2022), a public biomedical corpus, the same comparison raises AUPRC from 0.738 to 0.843 and entity-pair F1 from 57.5 to 67.1, with the gain where the argument predicts: absent on sentences that name only two entities, large on sentences that name more. Zero-shot Qwen models from 1.7B to 122B parameters show the same gap, and the verifier runs at 32 candidates per second on a CPU.

Contributions.

- A controlled comparison of sentence-level, pair-conditioned and triple-conditioned verification on one encoder and one corpus, replicated on a public benchmark and in zero-shot Qwen language models from 1.7B to 122B parameters, with the gain located where the argument predicts (§4).
- An account of what the pair query contributes, namely disambiguation between the organisms a passage names and a check on candi-

dates whose arguments are not organisms, and a bound showing that checking a triple’s components separately cannot reach the largest error class (§4.3, §4.6).

- A practical verifier: operating curves for choosing a threshold, deterministic candidate rules, a four-way direction label, CPU throughput, and six negative results (§4.7, §5, Appendix P).¹

2 Related Work

Partial-input baselines. That a classifier can score well by exploiting how its data were assembled is well established in NLP. Partial-input baselines recover much of the label from the hypothesis alone in natural language inference (Gururangan et al., 2018; Poliak et al., 2018) and from the passage alone in reading comprehension (Kaushik and Lipton, 2018), and such rules hold in distribution while failing under deployment (Geirhos et al., 2020; Du et al., 2024); the closest task-level precedent is fact verification (Thorne et al., 2018), where a claim-only classifier matches evidence-aware models because of how the corpus was built (Schuster et al., 2019). The sentence-level filter of §1 is a partial-input model in exactly this sense, except that the missing input is the query the pipeline itself issued rather than an annotator artefact (Wiegand et al., 2019; Parmar et al., 2023). As Feng et al. (2019) caution, our claims are limited to the models and corpora we study.

Entity pairs in relation extraction. Distant supervision assumes that a sentence containing two related entities expresses the relation (Mintz et al., 2009), and its standard relaxation exists because it does not (Riedel et al., 2010). Since TACRED an instance has been the triple (sentence, e_1 , e_2) rather than the sentence (Zhang et al., 2017), and how the pair is injected into the encoder is worth several F1 points (Baldini Soares et al., 2019; Wu and He, 2019; Zhou and Chen, 2022). Rosenman et al. (2020) already name our failure, as an “event heuristic” that classifies on (sentence, relation) while ignoring the arguments. Our contribution is a controlled measurement of what it costs in a verification setting built by co-occurrence retrieval, replicated on BioRED and in zero-shot Qwen models and located where the argument predicts; a bound on component-wise remedies; and the candidate

¹Code: <https://anonymous.4open.science/r/XXXX>.

rules. The entity-type shortcut usually invoked for such errors (Peng et al., 2020; Mtumbuka and Schockaert, 2024) is unavailable here, since both slots are filled from one inventory of taxon surface forms. Error attribution across pipeline stages is standard and easy to get wrong (Taillé et al., 2020; Ai and Kavuluru, 2023).

Conditioning on the query. Casting a relation as a question about a named entity (Levy et al., 2017), or as an entailment hypothesis verbalising the relation between two of them (Obamuyide and Vlachos, 2018; Sainz et al., 2021), is the reformulation our input format instantiates, and the resulting decision (does this passage support this specific claim?) has the shape of scientific claim verification (Wadden et al., 2020), and of validating a knowledge-graph triple against text (Lehmann et al., 2012), except that those systems retrieve evidence for a given triple, while here retrieval fixes the passage. Distilling a large model’s judgments into a small student is standard (West et al., 2022; Gekhman et al., 2023). Selective classification thresholds a score instead of issuing a verdict (Geifman and El-Yaniv, 2017), which is how extraction is made useful to human abstractors (Swaminathan et al., 2024).

Text mining for interaction databases. Judging an extracted fact before it reaches a curator is an old idea (Rodriguez-Esteban et al., 2006), as is evaluating such tools by whether they help the curator (Hirschman et al., 2012; Britan et al., 2018; Lee et al., 2018). Cuzick et al. (2023), independently, take the interacting pair rather than the passage as the annotation unit, the unit our pair-binding error class (§4.6) isolates. Large language models mine ecological text (Farrell et al., 2024; D’Souza et al., 2025) and extract interactions at scale (Keck et al., 2025; Zou et al., 2026); Zou et al. (2026) standardise names and categories post hoc, with no step that checks each extraction against its text. We are not aware of published work that verifies literature-mined biotic-interaction triples against their source passage before ingestion.

3 Method

3.1 Candidates and task definition

BiotXplorer (Ruch et al., 2024) proposes candidate biotic-interaction triples from the literature indexed by SIB Literature Services (Gobeill et al., 2020). For each passage it matches taxon surface forms against a taxonomy and interaction surface forms

against a vocabulary derived from the Relation Ontology, and emits candidate (s_1, r, s_2) triples with the passage, the matched surface strings and their canonical forms. We use it only as a source of candidates: no component of it is a baseline in this paper. The pool we study has 175,588 candidate rows over 165,325 distinct passages and 115,404 distinct triplet keys. Canonical strings often differ from what the passage says, so we always query with the surface form, which is what the model can see. The decision to make is whether the passage asserts an interaction between the candidate’s two organisms, in either direction (§3.5).

3.2 Training data and models

The teacher. Labels come from a locally hosted 32B open-weight instruction model (Qwen Team, 2025), prompted zero-shot at temperature 0 (prompts in Appendix A) and asked, per candidate, whether the passage supports this triple. Its labels therefore vary across candidates that share a passage. Against 150 expert-graded items its mislabel rate is $22/150 = 14.7\%$ (Clopper–Pearson [9.4%, 21.4%]; 10 false positives, 12 false negatives).

The corpus. 48,338 teacher-labelled rows, 0.343 positive, drawn from the same retrieval pool as the benchmark (disjoint passages): a 20,000-row base sample, 16,546 *contrast* rows, in which one passage and its pair recur with different relation terms (on 2,017 passages with labels that disagree), 6,201 actively sampled rows, and 5,591 rows with the candidate’s argument order reversed.

Models and inputs. All three are cross-encoders on BiomedBERT-base (Gu et al., 2021) (110M parameters), distilled from the teacher’s labels (Bucilua et al., 2006; Hinton et al., 2015) with the same recipe (Appendix O), keeping the checkpoint with the best development AUPRC on a 10% development split grouped by taxon pair so that no pair spans training and development. Three seeds each; one run takes about four minutes on a single A100. They differ only in their input. The **sentence-level** model reads the passage alone. It is the simplest formulation, and on rows that share a passage it is shown the same input with different labels, which it cannot fit. The **pair-conditioned** model reads the two candidate taxa, s_1 [SEP] s_2 , as segment A and the passage as segment B (Nogueira and Cho, 2019). The **triple-conditioned** model reads the whole candidate, s_1 [SEP] r [SEP] s_2 . Compar-

ing the three separates the effect of the pair from that of the relation term.

3.3 An expert-graded benchmark

One benchmark of 437 rows, 251 positive, in three blocks of different provenance; we report every block as well as the pooled figures, since pooling alone hides their disagreement (§4.3, Appendix C). BIOTX100 consists of retrieved candidates, each with one passage chosen by retrieval, expert-graded on four axes: whether each taxon surface form was correctly resolved, whether the relation was correct, and whether the whole triple is supported by its passage. Half were drawn from pairs already recorded in GloBI and half at random, and after removing three training duplicates it contributes 97 rows: 50 GloBI-matched with 43 positive, 47 random with 33 positive. REJECT50 consists of candidates an earlier filter rejected, expert-graded for the ones it should have kept: 41 clean rows, 11 positive. It is the paper’s recall-side instrument (Appendix I). TEST299 is 299 expert-graded pair-level items, 0.548 positive, and is where passages most often name several taxa.

The blocks do not carry the same gold: BIOTX100 alone is graded at the level of the whole triple, while TEST299 was expert-graded at the pair level and its stored relation term was attached by retrieval. We therefore evaluate everywhere at the level all three share (“do these two taxa interact in this passage?”), and use the triple-level grades only to diagnose (§4.6). This understates the benefit of triple conditioning, since a candidate with the right pair and the wrong relation counts as correct.

Two caveats apply. First, we scored roughly a hundred model configurations on this benchmark during development, so the recipes compared here were chosen with it in view; we expect the controlled comparison of §4 to be the least affected. Second, eight gold labels changed after the benchmark was frozen: seven by blind re-annotation of model-disputed items (Limitations), and a TEST299 candidate pairing *Acanthamoeba* with *Pseudomonas*, revised to negative after the expert, grading direction without model scores, recorded that the two co-occur in a shared host rather than interacting.

3.4 Replication setting: BioRED

The biodiversity benchmark is ours, mostly singly graded, and modest in size. To test the claim where anyone can check it, we rebuild the same setting

on BioRED (Luo et al., 2022), a public corpus of PubMed abstracts with expert-annotated entities and relations. Candidates are generated by co-occurrence: every pair of distinct annotated concepts co-mentioned in one sentence, restricted to the eight entity-type pairs that carry relations in training, with the sentence as passage. A candidate is positive when BioRED relates the two concepts anywhere in the abstract, a document-level label whose noise every model shares. We follow the protocol of Lai et al. (2025): models train on the 500 abstracts of BioRED’s training and development splits (15,640 candidates), select their epoch and threshold on its 100 test abstracts, and are scored on the 400 abstracts of the BioCreative VIII test set (Islamaj et al., 2024). BioRED has no retrieved relation term, so only the sentence-level and pair-conditioned formulations are compared, trained with the recipe of §3.2.

3.5 Evaluation protocol

The threshold policy is fixed in advance. A model’s reported operating point is either **pre-specified**, chosen on its own training-development split and never on the benchmark, or **block-held-out**: fitted on two of the three source blocks and applied to the third, so that every row is scored by a threshold that never saw it and every row is used. Comparisons between models use the same policy for both. Intervals are Clopper–Pearson; McNemar tests (McNemar, 1947) carry a continuity correction. With three seeds we report either the per-seed mean and standard deviation or a three-checkpoint ensemble, which averages the checkpoints’ probabilities. The benchmark ships an `in_train` flag on 9 rows; a 5-gram Jaccard scan against all 48,338 training rows finds 3 more at Jaccard 1.000, all in BIOTX100. All 12 are excluded throughout.

The gold is pair-level and therefore order-free, so a model that reads the candidate in order is scored as the higher of its two scores over the two argument orders, that is, the probability that the pair interacts in at least one direction. Otherwise a model would be penalised for rejecting a correctly paired candidate whose stated relation runs backwards, which the gold does not count as an error.

4 Results

4.1 Main results

Table 1 gives the comparison. Telling the model which pair a candidate is about raises per-seed

Model	Biodiversity (437)		BioRED (16,606)		
	AUPRC	F1	AUPRC	F1	EP F1
Accept all	0.574	0.730	0.463	0.633	–
Sentence-level	$0.867 \pm .006$	0.790	$0.738 \pm .004$	0.718	57.5
Pair-conditioned	$0.942 \pm .007$	0.882[†]	0.843$\pm .004$	0.817[†]	67.1
Triple-conditioned	0.945$\pm .005$	0.867 [†]	–	–	–

Table 1: Within each benchmark, all models share one encoder and one training corpus; only the input differs. AUPRC: mean $\pm$ s.d. over three seeds. F1: three-checkpoint ensemble at block-held-out thresholds (biodiversity) or the pre-specified threshold (BioRED). EP F1: BioRED’s document-level entity-pair F1 (%), seed mean. Accept all gives the base rate. [†]McNemar $p < 10^{-4}$ against the sentence-level model; pair-against triple-conditioned, $p = 0.26$. BioRED has no relation term, hence no triple-conditioned model.

AUPRC from 0.867 ± 0.006 to 0.942 ± 0.007 and, at block-held-out thresholds, F1 from 0.790 to 0.882 (precision 0.752 to 0.869, recall 0.833 to 0.896). On decision correctness the pair-conditioned model fixes 74 items and breaks 23, McNemar $p = 3.8 \times 10^{-7}$. The advantage does not depend on calibration: compared at equal review workload, it is ahead by 0.07 to 0.09 F1 whenever 50% to 70% of candidates are accepted, and the curves come within 0.01 F1 of each other only when fewer than 5% or more than 89% of candidates are accepted (Figure 2b; shares at six thresholds in Appendix C).

The triple-conditioned model may help where high precision is needed: at $\tau = 0.5$ it has precision 0.942 at recall 0.777, against 0.940 at 0.689 for the pair-conditioned model (Figure 2a; Appendix C), but it falls behind at the highest recall. Its AUPRC is level with the pair-conditioned model’s (0.945 against 0.942; 21 items fixed, 30 broken, $p = 0.26$). The re-annotation favoured this model most (Limitations), so we treat the difference as tentative.

4.2 Results on BioRED

The result (Table 1, right; full table in Appendix D) reproduces the biodiversity one. Conditioning on the pair raises AUPRC from 0.738 to 0.843 and F1 from 0.718 to 0.817, fixing 2,642 items and breaking 1,064. Marking the two mentions in the passage, the standard relation extraction input (Zhou and Chen, 2022), adds nothing to the pair query ($p = 0.08$).

On BioRED’s standard document-level metric the pair-conditioned model reaches an entity-pair F1 of 67.1, against 57.5 for the sentence-level one.

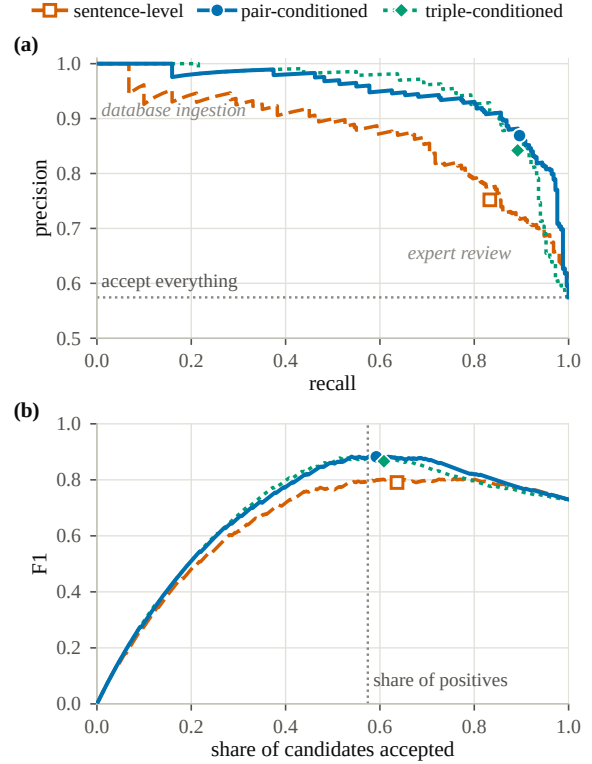


Figure 2: Operating curves on the 437 expert-graded candidates (three-checkpoint ensembles). (a) Precision against recall. Markers are the operating points of Table 1; their thresholds are fitted per block, so they can sit slightly off the pooled curve. Dotted line: accepting every candidate. (b) F1 against the share of candidates accepted, which compares the models at equal review workload whatever their calibration; the pair-conditioned model is ahead of or level with the sentence-level one at every share except 93% to 96%, where it trails by at most 0.003. Values at six thresholds: Appendix C.

Systems that read the whole abstract reach 74.1 to 75.3 (PubMedBERT, BioREx and BioREDirect; Lai et al., 2023, 2025) and zero-shot GPT-4 42.9 (Islamaj et al., 2024). Much of that gap is structural: a quarter of the related pairs are never co-mentioned in one sentence, which caps any sentence-scope verifier at recall 0.749. Given the whole abstract, the same model and recipe reach 67.0 (recall 0.697, precision 0.646). The published systems also classify entity pairs, so the remaining gap comes from their document-level modelling, which is a separate problem from ours.

4.3 Gain by number of taxa in the passage

Figure 3 locates the gain; it is the most direct test of the argument. On passages that name three or more taxa, adding the pair raises AUPRC by 0.098; on

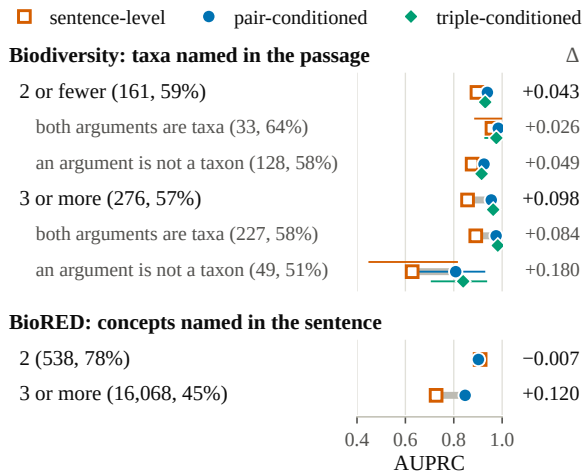


Figure 3: AUPRC by the number of entities the passage names (three-checkpoint ensembles). Biodiversity: distinct taxa that TaxoNERD (Le Guillaume and Thuiller, 2022) finds in the passage; an argument “is not a taxon” when TaxoNERD does not recognise it there. BioRED: gold-annotated concepts in the sentence; BioRED has no relation term, so there is no triple-conditioned model. Triple-conditioned markers sit just below their row so they do not hide the pair-conditioned ones. Grey bars and the Δ column give the gain from adding the pair (pair-conditioned minus sentence-level). Parentheses give the number of candidates and the share of positives. Thin lines on the rows under 100 candidates are bootstrap 95% intervals (2,000 resamples of candidates, fixed seed; values in `fig3_gain_by_taxa.json`). Exact values: Appendices C and D.

passages naming two or fewer taxa, by 0.043. Splitting by argument shows where this smaller gain comes from: the candidates come from automatic matching, and 41% of the 437 have one “taxon” that is a string TaxoNERD does not recognise as an organism in the passage (an organ, an author’s name, a vernacular term). Where both arguments are recognised taxa and the passage names no third one, the case in which the sentence question and the pair question nearly coincide, the pair query adds +0.026 on 33 items; where an argument is not a taxon it adds +0.049, and +0.180 when the passage also names others. The pair query therefore does two things. It tells the model which pair to judge, and it lets the model check that both arguments are organisms. BioRED, whose entities are gold-annotated, isolates the first: on candidates from sentences that name only two concepts the two models are indistinguishable (0.909 against 0.902; per seed 0.899–0.919 against 0.892–0.914), while on those from sentences naming more, which are 97% of the candidates, conditioning is worth

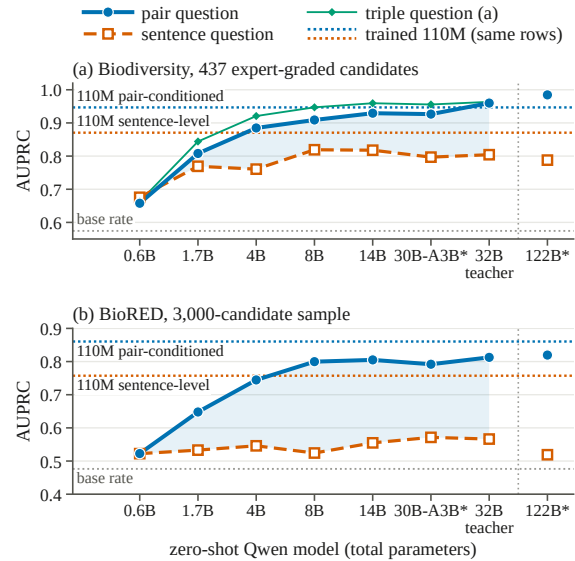


Figure 4: Zero-shot Qwen models (Qwen Team, 2025) asked the sentence question (dashed, open squares) or the pair question (solid, filled circles) on (a) the 437 expert-graded candidates and (b) a 3,000-candidate BioRED sample. Dotted: the trained 110M models on the same rows (three-checkpoint ensembles); grey: base rate. The x-axis spaces model sizes evenly. From 1.7B on, the pair question leads at every size, while answers to the sentence question change little with scale. *Mixture of experts, placed by total size; the 32B model is the teacher. Qwen3.5-122B-A10B is a later release. Thin green line in (a): the triple question (biodiversity only), which matches or exceeds the pair question (Appendix E).

+0.120. By source block, the pair query raises AUPRC from 0.835 to 0.944 on TEST299, whose passages name the most taxa, from 0.964 to 0.982 on BIOTX100, and from 0.551 to 0.575 on the 41-row REJECT50, too small to rank the models (Appendices C and I).

4.4 Zero-shot language models

The formulation effect is not a property of small encoders (Figure 4; scores are $P(\text{YES})$, Appendix A). Asked the sentence question, the 32B teacher scores AUPRC 0.804, below the 110M sentence-level student; asked about the pair, 0.960. Its gap, +0.156, is twice the student’s, and it sits in the same place: +0.179 on passages naming three or more taxa against +0.107 on the rest, where the student’s is +0.098 against +0.043. The 110M pair-conditioned student, about 300 times smaller than its teacher, scores 0.14 AUPRC higher than the teacher asked the sentence question. As with the student, the relation term adds nothing

at this size (0.964 against 0.960), although zero-shot models from 1.7B to 30B-A3B do gain from it (Appendix E). From Qwen3-1.7B to Qwen3.5-122B-A10B the pair question leads at each size, by +0.04 to +0.20 AUPRC on our benchmark and +0.11 to +0.30 on BioRED, while the sentence question gains little with size. On BioRED the teacher, asked the sentence question, accepts 92% of its sample (AUPRC 0.566 at a base rate of 0.48); asked about the pair, 0.813, still behind the trained pair-conditioned model (0.860 on the same sample). Against its teacher’s labelling prompt (0.964) the pair-conditioned student gives up 0.021 AUPRC, of which distilling the teacher’s probabilities recovers about a quarter (Appendix F). In F1 it gives up almost nothing at block-held-out thresholds (0.882 against the teacher’s 0.885; the soft-label student reaches 0.883).

4.5 Controls: query ablation, training data, encoder

Three controls locate the effect in the query. Degrading the trained triple-conditioned model’s query at inference, weights and thresholds unchanged, shows which part of the query it uses: deleting both taxa drops the three-checkpoint ensemble’s AUPRC from 0.948 to 0.691 against a base rate of 0.574, and emptying the query gives 0.649, while replacing or shuffling the relation, pair kept, moves it far less (0.906, 0.930; Appendix G). The conditioned model does not need the corpus’s contrast rows: a triple-conditioned model trained on the 20,000-row base sample alone reaches AUPRC 0.943 ± 0.004 (full corpus 0.945 ± 0.005) and block-held-out F1 0.886. The encoder is not the cause either: of seven alternative recipes screened on development data, the three that passed were scored once on test, none pays on both benchmarks, and with every encoder trained in both formulations the pair query wins, by +0.103 to +0.113 AUPRC on BioRED and +0.051 to +0.076 on ours (Appendix H).

4.6 Error analysis

The pair-conditioned model’s 34 false positives at its block-held-out operating point are not homogeneous: 24 come from passages naming three or more taxa, and 16 have an argument TaxoNERD does not recognise as a taxon. Read by hand, they sort into a few mechanisms (examples in Table 10, Appendix J; the per-row table is released): *pair-binding*, where both taxa and the relation are

correct but the relation’s argument is a third organism; *co-listing*, where the two taxa are members of one enumeration and the passage states no interaction between them; *negation*, where the passage asserts the opposite; *same organism*, where the two “taxa” are one organism named twice or an organism and its own clade; candidates whose relation term names no biotic interaction at all (*relative to*, *hybridization*, *migration*); and *entity* failures, where a taxon string is a taxonomic author, a place, an organ, or an adjective inside a pathogen’s name (“avian schistosomes”, which the pair-conditioned model already rejects). Some that read as gold errors were re-graded blind (Limitations).

A bound on component-wise verification. The pair-binding class limits a natural alternative architecture. Its members have both taxa correctly resolved and the correct relation; the error is a pairing made by retrieval that the passage does not state. A verifier that checks s_1 , r and s_2 separately, however good each check is, cannot see it. On the 97 clean BIOTX100 candidates, where triple-level grades make the classes countable, 15 of the 32 unsupported candidates are pair-binding failures (the rest: 9 entity, 8 predicate), a share of 0.47, Clopper–Pearson [0.29, 0.65]. A component-wise verifier accepting a fraction r of the 65 supported triples therefore also accepts all 15, so its precision is at most $65r/(65r + 15)$: 0.81 at full recall, [0.76, 0.88] across that interval. Only a model that reads the pair together with its passage can exceed this bound.

4.7 Candidate rules and a recommended configuration

Several of these mechanisms are questions about the candidate rather than the passage (whether two strings name one organism, whether a word is a surname, whether *relative to* is an interaction), and a classifier handles them poorly. We answer them with eight deterministic rejection rules applied before the verifier decides (Appendix L). No rule has a parameter fitted to evaluation labels. Since four were written after reading the false positives above, each had to pass a criterion fixed in advance on the 42,747 training rows in retrieved argument order (at least 85% of what it rejects is teacher-negative), and all were then frozen, recorded by hash and evaluated once. On the pair-conditioned model they buy 0.009 to 0.018 precision for 0.008 to 0.020 recall, F1 unchanged ($p = 1.00$): of its

34 false positives they catch 6, because a model told which pair it is judging already rejects most of what they target. On the triple-conditioned model they do more (precision $0.842 \rightarrow 0.883$, $p = 0.17$). Both are upper estimates, since half the rules were chosen with this benchmark’s errors in view; their main practical benefit is transparency, because each rejection names its rule.

The recommended configuration. The model we recommend lies outside the controlled comparison, differing from the pair-conditioned model of §3.2 in three respects adopted during development: the two taxa are given in alphabetical order, they are marked in place in the passage, and the training labels are the teacher’s pair-level judgments. Its threshold, 0.5, was set before evaluation. On the 437 candidates it reaches AUPRC 0.970 and F1 0.915 (precision 0.873, recall 0.960), or 0.917 with the candidate rules,² and scores 32.3 candidates per second on eight CPU threads. Escalating the 30% of candidates nearest its threshold to a zero-shot Qwen3.5-122B-A10B raises its precision with the rules from 0.888 to 0.962 at recall 0.912 (Appendix N). A second head labels the interaction’s direction in four categories; on 97 separately expert-graded candidates it answers 74% of the directional ones, 0.887 of those correctly, an optimistic figure since the abstention level was chosen on that set (Appendix M).

5 Discussion and Conclusion

Partial input and model size. A sentence-level filter is a partial-input model in the sense of §2, except that the input it lacks is the query the pipeline issued, so the remedy is to put that query into the input. Larger models do not make up for the missing input. From 1.7B to 122B, zero-shot Qwen models answer the pair question better at each size while their answers to the sentence question gain little, and a 110M model asked about the pair ranks candidates better than the 32B teacher asked about the passage (§4.4).

Passages naming only two taxa. The two questions coincide when a passage names only the candidate’s two entities and both are what they claim to be: on BioRED sentences that name two concepts the two models are indistinguishable, and on

²The *co_listed* rule is omitted from this configuration: on this model’s accept set it removes one false positive at the cost of three true ones.

our benchmark the gain is smallest where both arguments are recognised taxa and no third is named (§4.3). Most of our positives come from passages that name more.

Choosing an operating point. The pair-conditioned model’s precision ranges from 0.761 at $\tau = 0.01$ to 0.981 at $\tau = 0.90$ (Appendix C), so the right threshold depends on the use. Ingestion into a reference database calls for high precision, and assembling candidates for expert review calls for high recall. Reporting a score leaves this choice to the user (Geifman and El-Yaniv, 2017), and escalating uncertain candidates to a zero-shot LLM (§4.7) adds another option.

Errors made before verification. Errors such as a wrong interaction term or a mis-resolved taxon arise during retrieval, before any verifier runs, and a model that reads the passage cannot see that the resolved species is not the one the passage names (Limitations). Direction needs its own gold (Appendix K), and candidates drawn after the benchmark were harder for all three models.

Other domains. Any pipeline that proposes candidates by co-occurrence and filters them passage by passage gives every candidate from a passage the same answer; BioRED shows the remedy carrying over to genes, chemicals and diseases.

Conclusion. When candidates come from co-occurrence retrieval, the model that verifies them should see the candidate pair as well as the passage. A classifier that reads only the passage gives every pair in it the same answer, and 62% of the interacting pairs in our benchmark come from passages naming three or more taxa. Giving the model the pair raises AUPRC from 0.867 to 0.942, with the largest gain on those passages. The effect replicates on BioRED and in zero-shot Qwen models from 1.7B to 122B parameters. The verifier runs on a CPU, fast enough to score every candidate in the pool. For another pipeline, the first quantity to measure is the share of passages that name more than the candidate pair.

Limitations

Benchmark size and annotation. 437 expert-graded rows in three blocks, each graded once. A second annotation then re-graded 22 candidates blind, with labels and scores hidden: the 11 on which two students of the same teacher, the

triple-conditioned model and the recommended one (§4.7), both contradict the gold with confidence 0.9 or more, mixed with 11 random controls. All 11 controls kept their label (agreement 11/11, Clopper–Pearson [0.72, 1.00], too few to bound label noise tightly). Of the 11 contested labels, 7 changed, 6 to positive and 1 to negative; we adopted the re-annotation, and every benchmark number in this paper uses the revised gold except the historical rows of Table 14. The models chose which candidates were re-graded but changed no label; every change nonetheless moves the gold toward those two models, both of which read the candidate, and lifts their AUPRC most (+0.036 each, against +0.024 for the pair-conditioned and +0.015 for the sentence-level model), enough to bring the triple-conditioned model level with the pair-conditioned one. On the gold before re-annotation the pair query raises AUPRC from 0.851 to 0.918 and F1 from 0.775 to 0.871. One change concerns a passage stating that “the bat tick ... has been reported ... to occasionally feed on humans”, first labelled negative because retrieval had resolved the tick to the wrong species. Judged as written, the passage supports the pair, and the label is now positive; a database that ingested the resolved taxon would still record the wrong species, an upstream error no passage-reading model can see. On the recall side (REJECT50, Appendix I) the denominator is 11 positives.

Sampling of BIOTX100. BIOTX100 is a selected sample: each candidate comes with one passage chosen by retrieval, and half are pairs GloBI already records. Whether the gains hold on candidates and passages sampled without that selection, which are likely noisier, is not measured on this block; TEST299 and the public benchmark of §3.4 are sampled differently.

Rule selection. Four of the eight rules were selected after reading this benchmark’s false positives (§4.7). The training-data criterion bounds how much that selection can have memorised the benchmark, but it does not remove it, and their effects should be read as optimistic until confirmed on candidates neither the rules nor their author have seen. They fire on no row of a separate 100-candidate sample, so that sample can neither confirm nor contradict them.

Fresh candidates. On that sample (100 retrieved candidates drawn after the benchmark),

all three models score well below their benchmark figures, and the sentence-level model is again last: AUPRC 0.665 sentence-level, 0.785 pair-conditioned, 0.765 triple-conditioned. Its label provenance is not identical to the benchmark’s, so we do not read the drop as measured, but it is a sign that the benchmark may be easier than a live pool.

Pair-level evaluation. Only BIOTX100 carries whole-triple gold. Evaluating at the pair level everywhere means a candidate with the right pair and a misbound relation counts as correct, so the comparison cannot credit the relation term with the cases it alone could catch; triple-level gold on the other two blocks is what would test whether the triple-conditioned model’s advantage at strict operating points is real.

Document-level labels in BioRED. A candidate is positive when the abstract relates the two concepts anywhere, so a co-mention in a sentence that does not state the relation counts as positive. Both models are trained and scored on the same noisy labels, so the comparison is controlled, but the absolute figures understate what either would reach against sentence-level gold.

Teacher noise and pretraining exposure. The training labels come from a prompted teacher wrong about one time in seven, and the student inherits that boundary; training on model-generated labels can degrade encoders (Lu and Smith, 2025). The passages are published MEDLINE text and the teacher is a pretrained model, so we cannot exclude that it saw them, and its labels may be optimistic for genuinely new literature.

Upstream failures. Of the 41 REJECT50 candidates, 27 carry an incorrect interaction term and 12 a mis-resolved taxon, 10 of them both. Those fail before any verifier is consulted, and no change to the decision stage, ours included, addresses them.

Single retrieval system and domain. We study one retrieval system with one interaction vocabulary, on interactions between taxa in MEDLINE. The mechanism, a pair-agnostic decision over passages that name several organisms, is not specific to this pool, but we have not shown that other pools fail the same way.

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A Teacher prompts

The three questions put to the 32B teacher and the other zero-shot models (§4.4, Appendix E), verbatim, at temperature 0 with thinking disabled; the score is the probability of YES among the YES/NO alternatives of the first answer token. The triple question is the prompt that labelled the training corpus. On BioRED the sentence and pair questions name biomedical entities and relation types in place of organisms and biotic interactions (their exact text is released with the code). Lines are wrapped for the page.

Sentence question.

You check sentences retrieved from the scientific literature for a biotic-interaction database.

Sentence: <passage>

Answer YES only if the sentence asserts a direct biological interaction between organisms (for example predation, parasitism, infection, herbivory, pollination, mutualism, symbiosis or disease transmission), and the interaction is not negated.

Reply with exactly one word: YES or NO.

Pair question.

You verify candidate biotic-interaction pairs extracted from scientific literature.

Sentence: <passage>

Candidate pair:

Entity 1: "<taxon 1>"
Entity 2: "<taxon 2>"

Answer YES only if ALL of the following hold:

1. Both entities are genuinely the organisms referred to by those surface strings in this sentence (not a gene, protein, chemical, author name, or a different species).

seed	AUPRC			F1, $\tau = 0.5$		
	sent.	pair	triple	sent.	pair	triple
1	.871	.946	.948	.634	.804	.859
2	.868	.946	.947	.708	.772	.864
3	.860	.935	.939	.670	.802	.852
mean	.867	.942	.945	.671	.793	.858
sd	.006	.007	.005	.037	.018	.006

Table 2: Per-seed figures on the 437 biodiversity rows, order-free scoring. The body reports F1 of three-checkpoint ensembles at block-held-out thresholds; a fixed $\tau = 0.5$ is shown here only for the spread across seeds, not to compare the models, because the three models are calibrated differently, and a single shared threshold would favour one of them arbitrarily.

2. The sentence asserts a direct biological interaction between these two organisms (not merely a co-mention, and not an interaction each has with some third organism), and the interaction is not negated.

Reply with exactly one word: YES or NO.

Triple question (the labelling prompt).

You verify candidate biotic-interaction triples extracted from scientific literature.

Sentence: <passage>

Candidate triple:

Entity 1: "<taxon 1>"
Relation: "<relation>"
Entity 2: "<taxon 2>"

Answer YES only if ALL of the following hold:

1. Both entities are genuinely the organisms referred to by those surface strings in this sentence (not a gene, protein, chemical, author name, or a different species).
2. The sentence asserts a direct biological interaction between these two organisms (not merely a co-mention, and not an interaction each has with some third organism).
3. The stated relation is the correct type of interaction between them, in the stated direction, and is not negated.

Reply with exactly one word: YES or NO.

B Per-seed results

Table 2 gives the per-seed AUPRC behind the biodiversity means of Table 1.

τ	sentence				pair				triple			
	P	R	F1	Acc.	P	R	F1	Acc.	P	R	F1	Acc.
0.01	.709	.920	.801	.746	.761	.976	.855	.737	.740	.940	.828	.730
0.04	.782	.817	.799	.600	.815	.948	.877	.668	.811	.924	.864	.654
0.10	.819	.737	.776	.517	.854	.908	.880	.611	.845	.912	.877	.620
0.25	.861	.693	.768	.462	.911	.853	.881	.538	.903	.849	.875	.540
0.50	.886	.526	.660	.341	.940	.689	.795	.421	.942	.777	.852	.474
0.90	.930	.159	.272	.098	.981	.422	.591	.247	.982	.637	.773	.373

Table 3: Values behind Figure 2: the three models’ operating curves on all 437 rows, three-checkpoint ensembles; Acc. is the share of the 437 candidates the model accepts at τ . Rows show the shape; the reported operating points are those of Table 1.

Candidates	n	Pos.	Sentence	Pair	Triple
passage names ≤ 2 taxa	161	0.590	0.896	0.939	0.930
passage names ≥ 3 taxa	276	0.565	0.857	0.955	0.963
both arguments are taxa, ≤ 2	33	0.636	0.958	0.984	0.976
both arguments are taxa, ≥ 3	227	0.577	0.891	0.975	0.982
an argument is not a taxon, ≤ 2	128	0.578	0.876	0.925	0.915
an argument is not a taxon, ≥ 3	49	0.510	0.628	0.808	0.839
TEST299	299	0.548	0.835	0.944	0.951
BIOTX100	97	0.784	0.964	0.982	0.973
REJECT50	41	0.268	0.551	0.575	0.675

Table 4: Exact values of Figure 3 (biodiversity rows) and AUPRC by source block; all values are AUPRC of three-checkpoint ensembles, and Pos. is the share of positives in the row, the AUPRC of a random ranking. Taxa are counted as distinct mentions found in the passage by TaxoNERD (Le Guillarme and Thuiller, 2022) (en_ner_eco_biobert); an argument “is not a taxon” when TaxoNERD does not recognise it in the passage. TEST299 passages name 5.1 taxa on average, BIOTX100 2.6 and REJECT50 1.7.

C Values behind Figures 2 and 3

Table 3 gives the operating curves of Figure 2 at six thresholds, and Table 4 the AUPRC values of Figure 3, with AUPRC by source block.

D BioRED in full

Table 5 gives the BioRED comparison of §4.2 in full: AUPRC by the number of concepts the sentence names (§4.3), the entity-marker variant, and the zero-shot teacher on a 3,000-candidate sample (§4.4).

E Zero-shot scaling

Table 6 gives the zero-shot comparison of Figure 4 in full, with the triple question added. Each model’s score is $P(\text{YES})$ from its first answer token, and the triple question is the teacher’s labelling prompt, asked in both argument orders and maxed like the trained triple-conditioned model. At block-held-out thresholds the 32B teacher’s precision, recall

Model	AUPRC	F1	AUPRC, sentences naming		EP F1 (%)
			2 concepts	≥ 3	
Sentence-level	0.738 $\pm$.004	0.718	0.909	0.727	57.5
Pair-conditioned	0.843 $\pm$.004	0.817	0.902	0.847	67.1
+ entity markers	0.840 $\pm$.003	0.815	0.907	0.844	66.5
<i>Zero-shot 32B teacher, 3,000-candidate sample</i>					
Sentence question	0.566	0.662	0.921	0.555	n/a
Pair question	0.813	0.766	0.958	0.799	n/a

Table 5: BioRED, BioCreative VIII test set: 16,606 sentence-level candidates from 400 abstracts, 0.463 positive, 538 from sentences naming two concepts. Headline rows repeat Table 1. AUPRC is the per-seed mean $\pm$ standard deviation over three seeds (columns by concept count: three-checkpoint ensembles); F1 is at the pre-specified threshold (mean of the checkpoints’ development thresholds), the teacher’s at its greedy answer. EP F1 is BioRED’s document-level entity-pair F1, a pair scoring the maximum over the sentences that mention it. On the teacher’s sample the trained models’ three-checkpoint ensembles score 0.757 and 0.860. Pair-conditioned against sentence-level: 2,642 items fixed, 1,064 broken (McNemar $p = 5.9 \times 10^{-148}$).

and F1 are 0.654, 0.948 and 0.774 asked the sentence question, 0.919, 0.865 and 0.891 asked the pair question, and 0.878, 0.892 and 0.885 asked the triple question.

At 0.6B the three questions sit close to the base rate (0.66–0.68 against 0.574; 0.52 against 0.476): at that size the model barely does the task, whatever it is asked. From 1.7B on, the pair question is ahead at each size on both benchmarks: by +0.039 AUPRC at 1.7B and by +0.09 to +0.20 from 4B on (+0.11, then +0.20 to +0.30, on BioRED). The sentence question gains little with size (0.76–0.82 on ours from 4B on; 0.52–0.57 on BioRED throughout, near its base rate), so scale mostly improves the answer to the right question. The lead is not monotonic in size among the Qwen3 models, and it is largest for the 122B model: +0.197 on ours and +0.301 on BioRED, where the 32B model’s is +0.246. Asked the triple question, every Qwen3 model from 1.7B to 30B-A3B scores above its answer to the pair question on our benchmark, while the 0.6B, 32B and 122B models score level with it. Asked about the pair, the 122B model ranks our benchmark better than any trained model (0.985; the recommended model 0.970, the pair-conditioned student 0.942 per-seed mean, 0.947 as an ensemble), which suggests that a stronger teacher would raise the students further, whereas soft labels and larger encoders raise them little (Appendices F and H); asked the sentence question, it scores 0.788.

Qwen3	Biodiversity, 437				BioRED, 3,000		
	sent.	pair	triple	Δ	sent.	pair	Δ
0.6B	0.676	0.658	0.663	−0.018	0.522	0.523	+0.000
1.7B	0.769	0.808	0.844	+0.039	0.533	0.648	+0.115
4B	0.761	0.885	0.921	+0.124	0.546	0.745	+0.199
8B	0.819	0.909	0.947	+0.090	0.524	0.800	+0.275
14B	0.818	0.929	0.959	+0.112	0.555	0.805	+0.250
30B-A3B	0.797	0.927	0.956	+0.130	0.571	0.792	+0.221
32B (teacher)	0.804	0.960	0.964	+0.156	0.566	0.813	+0.246
Qwen3.5-122B-A10B	0.788	0.985	0.984	+0.197	0.519	0.820	+0.301
trained, same rows	0.871	0.947	0.948	+0.076	0.757	0.860	+0.103

Table 6: AUPRC of zero-shot Qwen3 (Qwen Team, 2025) by size, asked the three questions of Appendix A (4-bit builds served locally, thinking disabled; the triple question in both argument orders, maxed). 30B-A3B is a mixture of experts with 3B active parameters; Qwen3.5-122B-A10B (Qwen Team, 2026), a later release, is a mixture of experts with 8 of 256 experts active per token, run partly on the CPU. Base rates 0.574 and 0.476. Last row: the trained three-checkpoint ensembles on the same rows. Δ : pair minus sentence question (last row: pair-conditioned minus sentence-level model).

F Soft-label distillation

The students learn from the teacher’s verdicts. We re-queried the teacher for $P(\text{YES})$ on all 48,338 training rows (same prompt and inputs; its verdicts agree with the corpus labels on 99.7% of rows, and 24% of the probabilities lie between 0.05 and 0.95) and trained the three models again against these soft targets with a soft cross-entropy, selecting checkpoints on the same hard-labelled development split. It is the only distillation variant we tried, scored once (Table 7). Every model gains a little AUPRC (+0.005 to +0.007), the triple-conditioned model on every seed and with 0.021 F1 ($p = 0.03$ before correcting for three comparisons); against the pair-conditioned model it stays marginally ahead (0.951 against 0.947; 25 items fixed, 21 broken, $p = 0.66$). The comparison is unchanged: the pair query lifts AUPRC from 0.873 to 0.947 (72 fixed, 28 broken, $p = 1.7 \times 10^{-5}$), by +0.095 on passages naming three or more taxa and +0.036 on the rest. Probabilities close about a quarter of the pair-conditioned model’s distance to its teacher’s 0.964 (0.016 AUPRC, against 0.021 with verdicts). The body keeps the verdict-trained models, on which every other analysis was run.

G Query ablation

Degrading the trained triple-conditioned model’s query at inference, with weights and thresholds unchanged, shows which part of the query it uses (Ta-

Model	Target	AUPRC	F1	AUPRC, taxa	
				≤ 2	≥ 3
Sentence	verdict	0.867 $\pm$.006	0.790	0.896	0.857
	$P(\text{YES})$	0.873 $\pm$.007	0.799	0.908	0.862
Pair	verdict	0.942 $\pm$.007	0.882	0.939	0.955
	$P(\text{YES})$	0.947 $\pm$.003	0.883	0.944	0.957
Triple	verdict	0.945 $\pm$.005	0.867	0.930	0.963
	$P(\text{YES})$	0.951 $\pm$.003	0.888	0.936	0.965

Table 7: The sentence-level, pair-conditioned and triple-conditioned models trained on the teacher’s verdicts (the body’s models) and on its probabilities, recipe and seeds unchanged. AUPRC is the per-seed mean $\pm$ standard deviation; F1 (block-held-out thresholds) and AUPRC by the number of taxa the passage names are those of three-checkpoint ensembles, as in Tables 1 and 4. Per-seed AUPRC with soft targets: .866/.879/.875, .951/.946/.945 and .952/.952/.948. Against the verdict-trained model of the same format, the soft-target model fixes 13 items and breaks 7 (sentence-level), 13 and 14 (pair-conditioned), 19 and 7 (triple-conditioned; McNemar $p = 0.26, 1.0, 0.03$).

Query given to the model	P	R	F1	AUPRC
Retrieved triple	0.842	0.892	0.867	0.948
Generic relation, pair kept	0.918	0.673	0.777	0.906
Relation shuffled across items	0.900	0.721	0.801	0.930
Relation only, no taxa	0.833	0.020	0.039	0.691
Empty query	1.000	0.036	0.069	0.649

Table 8: Degrading segment A of the trained triple-conditioned model at inference, weights and its own block-held-out thresholds unchanged; AUPRC of the three-checkpoint ensemble. Base rate 0.574. Per-seed standard deviations on AUPRC are ≤ 0.032 .

ble 8; §4.5). Deleting both taxa collapses ranking from 0.948 to 0.691 against a base rate of 0.574, and emptying the query entirely gives 0.649; replacing or shuffling the relation, pair kept, moves ranking far less (0.906, 0.930): the model leans on the pair, which is what the pair-conditioned model already showed.

The empty-query variant is not the sentence-level model: deprived of the query it was trained with, it does far worse (0.649) than a model trained without one (0.867); hence three trained models.

H Encoders

Besides the recipe of §3.2 we tried seven others on development data only: five epochs at learning rates 1, 2 and 3×10^{-5} , BiomedBERT-large at 1 and 2×10^{-5} , and BioLinkBERT-base and -large, pretrained on PubMed with citation links (Yasunaga et al., 2022). The three that beat the base recipe by 0.01 development AUPRC or more on

Encoder	BioRED, BC8 test			Biodiversity, 437			
	sent.	pair	EP F1 (%)	sent.	pair	triple	pair F1
BiomedBERT-base	0.738	0.843	57.5/67.1	0.867	0.942	0.945	0.882
BiomedBERT-large	0.741	0.843	57.6/67.4	n/a	n/a	n/a	n/a
BioLinkBERT-base	0.743	0.855	57.9/67.9	0.888	0.939	0.949	0.860
BioLinkBERT-large	0.737	0.848	56.1/68.2	0.881	0.949	0.951	0.896

Table 9: The three formulations under the encoders that were scored on test, recipe otherwise that of §3.2: AUPRC and EP F1 (sentence-level/pair-conditioned) as per-seed means over three seeds, and block-held-out F1 of three-checkpoint ensembles. BiomedBERT-large was not trained on the biodiversity arms (Appendix P). Against the base encoder’s model of the same format, BioLinkBERT’s pair-conditioned model fixes 621 BioRED items and breaks 537 (McNemar $p = 0.015$) and on ours 19 and 32 ($p = 0.093$); its sentence-level model on ours, 42 and 20 ($p = 0.0077$). No BioLinkBERT-large model differs significantly from the base encoder’s on ours ($p \geq 0.23$).

both splits (seed 1, pair-conditioned model) were scored once on the test sets: BiomedBERT-large at 2×10^{-5} (Appendix P), and BioLinkBERT in its base size under the base recipe and in its large size (below). BioLinkBERT-base’s development AUPRC is 0.910 for the BioRED pair-conditioned model over three seeds, against 0.892 for the base encoder and 0.905 for BiomedBERT-large, and 0.852 for the biodiversity pair-conditioned model on seed 1, then 0.843 and 0.833 on seeds 2 and 3. Its BioRED pair-conditioned model gains on every seed (AUPRC 0.853–0.858 against 0.839–0.847; EP F1 67.7–68.0 against 66.8–67.5), the best sentence-scope AUPRC we have on BioRED (Table 9). On our benchmark its pair-conditioned model does not gain (0.939 ± 0.004 against 0.942 ± 0.007 ; F1 0.860 against 0.882) while its sentence-level model does (0.888, F1 0.816 against 0.790), so the formulation gap narrows to +0.051 AUPRC (53 items fixed, 37 broken, $p = 0.11$), still concentrated on passages naming three or more taxa (+0.073 against +0.009). On BioRED it widens, to +0.113.

BioLinkBERT-large, at 2×10^{-5} for three epochs like BiomedBERT-large, passed the same screen (development AUPRC 0.907 on BioRED and 0.869 on ours, seed 1; three-seed means 0.903 and 0.862). Its pair-conditioned models gain a little on both benchmarks: on BioRED, AUPRC 0.848 ± 0.009 and EP F1 68.2 ± 0.7 (seeds 67.7–68.9, each above every base seed); on ours, 0.949 ± 0.003 and F1 0.896. Neither gain is significant: against the base encoder’s pair-conditioned model it fixes 652

BioRED items and breaks 653 ($p = 1.0$), and on ours 28 and 22 ($p = 0.48$). Its other models, trained afterwards and scored once, complete the comparison: the sentence-level models reach 0.881 on ours and 0.737 on BioRED (EP F1 56.1), the triple-conditioned model 0.951, and the pair query again wins, by +0.068 on ours (74 items fixed, 28 broken, $p = 8 \times 10^{-6}$; +0.090 on passages naming three or more taxa, +0.026 on the rest) and by +0.111 on BioRED. We keep the base encoder: BioLinkBERT-base does not carry to our pair-conditioned model, and BioLinkBERT-large’s gains are not significant on either benchmark.

I Recall on rejected candidates

REJECT50 asks whether candidates an earlier filter rejected contain recoverable positives, and it is too small to rank the models: eleven positives, so one item moves recall by nine points. At block-held-out thresholds the sentence-level model recovers 9 of the 11 by re-admitting 13 of the 30 correct rejections (F1 0.545), the pair-conditioned model recovers 6 while re-admitting 5 (0.545), and the triple-conditioned model recovers 8 while re-admitting 3 (0.727). We read only the qualitative point: the discarded candidates do contain recoverable interactions, and a model that reads the candidate finds them at a lower cost in wrong re-admissions than one that reads only the passage.

J Error examples

The examples illustrate the mechanisms of §4.6, one each (Table 10).

K Entity disambiguation and direction

Taxonomy gate. The natural remedy for entity errors is to gate candidates on taxonomic validity. Against the Open Tree Taxonomy (Rees and Cranston, 2017) such a gate catches 1 of the 32 unsupported BIOTX100 candidates and discards 5 of the 65 supported ones. This is because *Illiger*, *Dikarya*, *copia* and *sulla*, the surnames, clades, transposon families and vernacular names behind the entity failures, all resolve in the taxonomy. The failure is therefore one of disambiguation, of the kind species-name normalisers address (Gerner et al., 2010), which a validity gate cannot detect; and the gate’s casualties are viruses, which are legitimate partners in this domain. The *authority* and *same_organism* rules (§4.7) target the disambiguation directly and are the better instrument.

Mechanism	Candidate triple	Passage (abridged)
Pair-binding	(<i>T. gondii</i> , parasite of, swine)	“The entero-pathogenic <i>Cystoisospora suis</i> , a coccidian parasite of swine and close relative to <i>Toxoplasma gondii</i> ...”
Co-listing	(katyids, prey, beetles)	“... remains were from insects ... (e.g. beetles , dragonflies, cockroaches, and female katyids), indicating that eavesdropping on prey signals is not the only prey-finding strategy used by this <i>bat</i> .”
Negation	(wheat, hosts, <i>F. acuminatum</i>)	“... <i>F. acuminatum</i> in New Zealand was found only on non-wheat hosts .”
Same organism	(<i>P. chinensis</i> , pollinators, Chinese knotweed)	“... potential pollinators of Chinese knotweed (<i>Persicaria chinensis</i>) .”
Entity	(Siberia , migrating, birds)	“... a portion of birds migrating from Siberia have common winter nesting areas.”

Table 10: One example of each mechanism among the pair-conditioned model’s false positives. The relation’s true argument is italicised where it is not the queried taxon; in the entity case a place name was matched as a taxon.

System	P	R	F1	fixed	broken
Pair-conditioned	0.869	0.896	0.882	n/a	n/a
+ 7 rules	0.878	0.888	0.883	3	2
+ 8 rules	0.887	0.876	0.882	6	5
Triple-conditioned	0.842	0.892	0.867	n/a	n/a
+ 7 rules	0.860	0.880	0.870	6	3
+ 8 rules	0.883	0.869	0.876	13	6

Table 11: Deterministic candidate rules applied before the model’s decision, all 437 rows, three-checkpoint ensembles at block-held-out thresholds. “Fixed” and “broken” are relative to the same model without rules. McNemar: $p = 1.00$ and 1.00 on the pair-conditioned model, 0.50 and 0.17 on the triple-conditioned model.

Direction labels. The benchmark’s expert grading marks reversed triples as correct; we verified two by hand, (*Breinlia*, host of, *Rattus*) and (*Demi-dospermus*, parasitized by, *A. inermis*). No directional number on the benchmark would therefore measure the model rather than the convention, which is why direction is scored on its own annotation (Appendix M). The supervision itself is also uneven: among the 5,591 reversed training rows, the rate at which the teacher rejects a reversal is governed by the relation’s surface string. Direction-encoding participles and prepositional forms are rejected at 0.51–0.70 (*affecting* 0.70, *affected by* 0.61, *parasite of* 0.57), bare relational nouns at 0.04–0.10 (*host of* 0.08, *pollinators* 0.04). Whether a training label encodes direction therefore depends on which string retrieval matched.

L Candidate rules in detail

The eight rules of §4.7 are applied to the candidate before the verifier decides, and each targets an error mechanism of §4.6. *same_organism* (aposition of a common and a scientific name, or

identical forms), *own_clade* (an adjacent clade annotation such as “(Acari: Prostigmata)”), and *authority* (a taxonomic author parsed as a taxon) detect the same-organism and entity classes. *non_biotic_relation* rejects phylogenetic, comparative and biogeographic relation terms; *negated* rejects the two negation forms whose scope provably contains one of the pair (“non-*s* hosts”, “no infections with ... *s*”); *not_an_organism* rejects organs and syndromes; *pathogen_modifier* rejects an adjective that only ever occurs inside a pathogen’s name; and *co_listed* rejects two taxa that are conjuncts of one dependency-parsed list not governed by an interaction noun (“interactions between *A* and *B*” is kept). Table 11 gives their effect on both models.

Validation. No rule has a parameter fitted to evaluation labels, but rules can still be fitted by choice, and four of these (*non_biotic_relation*, *negated*, *not_an_organism* and *pathogen_modifier*) were written after reading the false positives of §4.6. To keep their selection from simply memorising the benchmark, every rule had to pass a criterion fixed before it was evaluated: on the 42,747 training rows, at least 85% of the candidates it rejects must be teacher-negative, matching the teacher’s own accuracy. Two first drafts failed it and were narrowed on training data alone (a general negation cue fired on “without affecting the host”, and a role-noun list rejected “viruses” and “fungi”, which are legitimate coarse taxa), and the final rules reject teacher-negatives at 0.88 to 1.00 (*co_listed*, which needs a dependency parse, on a fixed 6,000-row sample). The rules were then frozen, recorded by hash, and evaluated once.

Expert-graded label	<i>n</i>	answered	correct
forward or reverse	84	62 (74%)	55 of 62 (0.887)
undecidable (“?”)	12	n/a	9 left UNCERTAIN
bidirectional	1	n/a	1 BIDIRECTIONAL

Table 12: The recommended model’s four-way direction output against 97 expert-graded direction labels. “Answered” counts FORWARD or REVERSE; the rest are UNCERTAIN. The abstention level (0.60) was chosen by the expert after seeing this set’s coverage curve, so the figures are optimistic.

M Direction head

The recommended model of §4.7 is one 110M-parameter model with two heads. It gives the two taxa in alphabetical order, so the input is identical whichever way round retrieval presents the pair, and marks them in place in the passage (@...@, #...#) rather than only naming them in segment A. Its interaction threshold was never tuned; with the candidate rules it reaches precision 0.888, recall 0.948 and F1 0.917 (5 items fixed, 3 broken, $p = 0.72$), and its CPU throughput is measured with both heads.

The second head reports the interaction’s *direction* in four categories. FORWARD and REVERSE say which taxon acts on the other; they come from a direction score that is exactly antisymmetric in the two marked taxa, and are reported only when it is confident, otherwise the answer is UNCERTAIN. BIDIRECTIONAL is assigned when the relation is mutual by definition (mutualism, symbiosis, competition, co-infection, *interacts with*), as listed in a relation-polarity lexicon derived from the Relation Ontology; for those, neither taxon is the subject and the model is not asked. The benchmark’s expert grading scores reversed triples as correct (Appendix K), so direction is evaluated on 97 separately expert-graded candidates (Table 12): the model answers 74% of the directional ones, correctly on 0.887 of those.

N Escalating uncertain candidates to the teacher

The recommended model is cheap and recall-oriented, and the zero-shot teacher, asked about the pair, is precise but slow (Appendix E). Table 13 combines them: the model decides every candidate, and those it is least sure of (a fixed fraction, nearest its threshold) are re-decided by a zero-shot LLM. Escalating 30% to Qwen3.5-122B-A10B raises precision from 0.888 to 0.962 at recall 0.912, remov-

Band	LLM calls	Qwen3-32B			Qwen3.5-122B-A10B		
		P	R	F1	P	R	F1
0%	0	0.888	0.948	0.917	0.888	0.948	0.917
10%	44	0.908	0.940	0.924	0.912	0.944	0.928
20%	87	0.920	0.920	0.920	0.932	0.924	0.928
30%	131	0.950	0.912	0.931	0.962	0.912	0.937
50%	218	0.964	0.849	0.903	0.981	0.841	0.906
100%	437	0.960	0.773	0.857	0.984	0.757	0.856

Table 13: The recommended model with its candidate rules keeps its own decision except on the band of candidates whose score lies nearest its threshold, which take a zero-shot LLM’s answer to the pair question (Qwen3-32B, the teacher, or Qwen3.5-122B-A10B); 437 rows. A rule rejection is final. Band 0% is the recommended model alone, 100% the LLM alone (with the rules).

ing 21 of the 30 false positives; the model alone reaches that precision only at $\tau = 0.99$, where recall falls to 0.745. Against the model alone the decisions are not significantly better on these 437 rows (25 items fixed, 13 broken, McNemar $p = 0.07$; with Qwen3-32B, 22 and 13, $p = 0.18$): the gain lies in the operating point, with more recall at a given precision, while accuracy at the model’s own threshold does not improve significantly. We report the band at six settings and do not choose one. Its cost is set by the teacher’s speed: escalating 30% of the 175,588-candidate pool means about 53,000 calls, under a day on one A100 for Qwen3.5-122B-A10B (1.4 s per call with part of it on the CPU) and less for the 32B.

O Artifacts and compute

The training recipe of §3.2 is AdamW at learning rate 2×10^{-5} and weight decay 0.01, a linear schedule with 10% warm-up, batch size 32, inputs of at most 256 wordpieces, three epochs and gradient clipping at 1.0; Appendix H lists the alternatives we screened. The encoders are BiomedBERT-base (microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext) and -large (MIT licence) and BioLinkBERT-base and -large (Apache 2.0); the zero-shot models are Qwen3 0.6B-32B and Qwen3.5-122B-A10B (Qwen Team, 2026) (Apache 2.0), run locally through Ollama; taxon mentions are counted with TaxoNERD (MIT). BioRED and its BioREDirect directionality release are public NCBI resources, used as distributed. Everything ran on one NVIDIA A100 (80 GB): an encoder run takes between two and fifteen minutes, and the whole study took about 50 GPU-hours,

most of them LLM inference for labelling the corpus and for the zero-shot rows.

P Negative results and cost

Intervention	Benefit	Cost
Taxon-validity gate	1 of 32 unsupported caught	−5/65 supported; rejects viruses
Self-distillation [†]	none	−0.001 P, −0.008 F1
6-layer compression [†]	1.9× CPU throughput	−0.088 precision
Corpus signal-matching [†]	none	−0.059 AUPRC
Rule-family training data [†]	none	AUPRC +0.005 to −0.037
BiomedBERT-large	+0.013 dev AUPRC (BioRED pair)	none on test; 3× parameters

Table 14: Six plausible design choices that measurably do not pay. [†]Measured under an earlier version of the evaluation; deltas are directions only (see text). The last row compares per-seed means.

Table 14 collects interventions that did not help, because each is a plausible first thing to try. Self-distillation at equal capacity is null, suggesting the teacher’s knowledge is already saturated in the student. Compressing to six layers nearly doubles CPU throughput, which this setting does not need, and costs 0.088 precision. Reweighting the corpus toward the signal distribution of the retrieval pool cost 0.059 AUPRC. The fifth row is the most informative. before the candidate rules existed, we turned the same mechanisms into training data (synthetic negatives for the same-organism, own-clade and authority classes, and in a second variant for co-listing) and retrained. The first variant left AUPRC unchanged within seed noise and the second cost 0.037. For properties of the candidate itself, rules worked better than extra training data. Deltas in rows two to five were measured under an earlier version of the evaluation and are given as directions. The last row is one of the three recipes of §4.5 that passed the development screen (the others are in Appendix H): BiomedBERT-large raises the BioRED pair-conditioned model’s development AUPRC from 0.892 to 0.905 (three-seed means), but on the BC8 test set the pair-conditioned model scores 0.843 ± 0.005 against 0.843 ± 0.004 , F1 0.814 against 0.817 , EP F1 67.4 ± 0.4 against 67.1 ± 0.3 , and the sentence-level model 0.741 against 0.738 ; the pair query’s gain

is unchanged ($+0.103$ against $+0.105$ AUPRC). It was not trained on all biodiversity arms, since its BioRED test result could not change the choice. For the whole-abstract BioRED model of §4.2 we also tried, on development data only (seed 1), eight epochs instead of three, alone and with each concept’s BioRED type in the query: development AUPRC 0.787 and 0.791 against 0.786, inside the spread of the three-epoch seeds (0.765–0.786), so neither was trained further or scored on the test set.